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ON THE MYTH OF RISING INTONATION IN POLAR QUESTIONS

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Using an extensive corpus of conversational data, it is shown that the role of Rising intonation (i.e. Rises, Fall–Rises, and Fall + Rises) in polar questions is overrated. Two types of polar – or 'yes/no' – questions are investigated: Inversion-questions (e.g. *Is this a question?*), and Queclaratives (e.g. *This is a question?*). In Inversion-questions, though Rising intonation is relatively frequent, the most frequent tone, in absolute terms, is a Fall; moreover, intonation is not used to distinguish genuine Inversion-questions from interrogatives without Question-status, such as Rhetorical Questions and Requests. In Queclaratives, a Falling intonation contour is by far the most frequent pattern, mostly accompanied by a step-up in pitch in the Head of the Tone Unit. Attention is also paid to the Pitch Range of polar questions, and to Pausal aspects of Question – Answer pairs. In all, the claim that Rising intonation (and, more particularly, final Rises) is the 'normal' pattern for polar questions lacks empirical justification.

1. Preliminaries

1.1. Rising intonations and polar questions

Within the 'British tradition' of prosodic description, one comes across statements such as the following, regarding the intonation of polar (or 'yes/no'-) questions:

"This [the low bounce, i.e. a contour ending in a low rise] is by far the most common way of asking such questions; it should be regarded as the normal way (...)" (O'Connor and Arnold (1973: 55))

Questions are thus assumed to end typically with a rising pitch movement; the same is true for Queclaratives (i.e. utterances with declarative form functioning as questions):

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"[C]omplete statements said with the high bounce [a high rising contour] have the effect of questions in most cases." (O'Connor and Arnold (1973: 75))

Similar claims are found in Kingdom (1958: 210) and Halliday (1968: 59), the implication being that rising tones are somehow 'unmarked' on polar questions.

Even the most cursory glance at conversational data shows this claim to be exaggerated: questions occur with a variety of contours (a fact which is acknowledged by most authors). Of course, one could argue that questions with a rising tone are more 'neutral' in meaning than those with other tones; however, it is not always easy to decide which meaning is the most neutral one (a fact which is also pointed out by Cruttenden (1986: 96)). One way out is to regard the most frequent tone as the basic one. We will try to show here that, on this basis, there is no justification for regarding rising contours as the unmarked pattern for polar questions. A study by Fries (1964) seems to support this point of view, as it concludes:

"The quantitative information does not support the much repeated assertion that yes/no questions 'regularly', 'usually', 'characteristically' have a rising intonation pattern." (Fries (1964: 250))

It should be pointed out, however, that Fries' study deals with American English rather than British English.

Before looking at the data, we have to settle some terminological matters. The term *Interrogative* will be used as a purely formal label; only utterances functioning as genuine requests for information will be termed *Questions* (see the following section for a discussion of the problems involved in assigning Question-status to an utterance). Two types of polar question will be investigated: Inversion-questions (with SV-inversion), and Queclaratives (cf. sections 2 and 3, respectively). The former category per definition also has Interrogative status (by virtue of the inversion), but this does not hold the other way round: Not every Inversion-interrogative is a genuine question. This problem does not arise for Queclaratives, which are a combination of non-interrogative form and question function.

1.2. Criteria for question identification

There is no simple correlation between Interrogative form and Question status. Some questions do not have interrogative form (i.e. Queclaratives); conversely, some interrogatives do not function as questions. We thus need other criteria for determining whether an utterance is a question or not.

Searle (1976) characterizes Questions as Directive speech acts, i.e. "attempts by (...) the speaker to get the hearer to do something" (Searle (1976: 11)). More specifically, they are attempts to get the hearer to supply a piece of verbal information. Searle (1969: 66–67) goes on to formulate a number of

criteria or 'felicity conditions' under which an utterance qualifies as a question. However, as has been pointed out by Levinson (1983: 289 ff.), there is no principled way of mapping speech acts onto utterance types; Searle's criteria, therefore, have no practical value for our purposes.

Are we then stuck with complete arbitrariness, as far as assigning question status to utterances is concerned? Fortunately, no. Participants engaged in conversation seem to have no problems whatsoever in identifying questions and supplying appropriate replies to them. In transcripts of conversations, the analyst has access not only to the questions asked by a speaker, but also to the hearer's/hearers' interpretation. In other words, we have at our disposal *both* parts of the Question–Answer 'adjacency pair' (Sacks et al. (1974)). If we can show that an utterance functions as the first half of such a Question–Answer pair, we will have arrived at an operational criterion for identifying Questions.

Polar questions, as the name suggests, require the hearer to make a positive/negative evaluation of the state of affairs expressed in the Question; a typical Answer would be one of the following three possibilities:
- confirmation (e.g. *yes*);
- rejection (e.g. *no*);
- statement to the effect that confirmation/rejection cannot be provided (e.g. *perhaps*; *I don't know*).

Since utterances such as the above do not make sense except as responses to Questions, they can safely be regarded as the second half of an adjacency pair; hence the other half must be a Question. Note that this procedure is independent of the actual form of the utterance. If an interrogative fails to evoke a response, chances are that the hearer (and hence the analyst) does not identify it as being a Question; the only other explanation would be that the hearer is uncooperative and violates Grice's (1975) first maxim of Quantity.

However, so far we have only dealt with the 'Unmarked' type of Question–Answer pairs; we must also allow for more 'Marked' instances such as (1) and (2) below:

(1) A: do you think this one's better?
B: do you mean the one on the left?

B's utterance hardly qualifies as a 'typical' response; nevertheless, A is clearly a question. Consider also (2):

(2) A: do you want a beer?
B: does a bear shit in the woods?

Unlike in (1), B does provide an answer to A's question (more specifically, a confirmative one); however, the answer is 'Marked' in that the response is

implicated rather than stated in a direct manner. The response in (2), which is near-idiomatic in American English, also carries the additional implicature that B finds it obvious that his answer should be confirmative.

Marked responses are problematic, in that they might once again lead to arbitrariness: by regarding every turn-change as a potential Marked answer to a question. Rather than try and formulate precise criteria for Marked responses, we will employ the following practical procedure for identifying Questions in our corpus, a procedure which is a blend of formal and functional criteria (the latter being more important):

(3) *Question-identification procedure*:
- (a) criterion A: for every turn-final utterance U, assume potential Question status;
- (b) criterion B: if U is Interrogative, and followed by an Unmarked response, accept Question status; if not, look for Marked response. If found, also accept Question status of U;
- (c) criterion C: if U is non-Interrogative, and followed by an Unmarked response, accept Question status; if not, reject U as Question.

This procedure rests on the assumption that in most cases form and function will coincide; when this is not the case (e.g. Queclaratives), criterion C comes into play. This criterion has a conservative bias, in that Marked responses are not accepted; thus, we can be reasonably sure that the remaining utterances will be genuine Questions, more specifically Queclaratives.[1]

It need hardly be said that the procedure in (3) is a purely practical device. We will attach no theoretical importance to it, and we definitely do not wish to imply that hearers follow a similar procedure for recognizing Questions in conversation.

Two problems remain. The first has to do with the adjacency requirement for Questions and Answers, which is implicit in (3). Consider exchanges (4) and (5):

(4) A: do you want to go to the movies (Q1)
B: you mean right now (Q2)
A: yes (A2)
B: no sorry I can't leave just now (A1)

[1] It should be pointed out that the procedure is actually a bit too powerful, in that some apparent, unmarked responses are not really responses at all. Consider, for instance, the following exchange:
– A: John told me he isn't coming.
– B: Yes, so I've heard.
Speaker A's utterance is clearly not a question here. It would appear that the only way to eliminate such instances is by using commonsense reasoning (we are indebted to the Journal of Pragmatics referee for pointing this specific problem out to us).

(5) A: do you like this painting? I hate it.
B: I think it's great actually

These instances show that it would be wrong to expect strict adjacency between Question and Answer all the time (see also Levinson (1983:304)). In (4) Q1 and A1 clearly form a Question–Answer pair, A1 being the Unmarked response to Q1; however, they are not adjacent, as there is another adjacency pair intervening. In (5), A's question is not even turn-final, as he adds some comment before allowing B to reply. Clearly, allowance must be made for 'near-adjacent' pairs such as these. Question and Answer may even partially overlap, as we will see later on.

The other problem concerns so-called Indirect Speech Acts (Searle (1975)) such as the following:

(6) A: could you pass me the rice please
B: oh yeah, sure (passes rice)

The first utterance is clearly a Request, asking for a non-verbal response from B. However, B also gives a verbal response; if we apply criterion (3b), his utterance qualifies as a Question. This implies that utterances can somehow have two illocutionary forces at the same time. We see no reason to reject this possibility, though we will not go into all its theoretical implications. In (6), the Request aspect is clearly the more important one, the verbal response merely adding a touch of politeness on B's part; in some cases, however, where there is real uncertainty about the hearer's willingness and/or ability to perform the requested action, the Question aspect may be equally important. We will keep instances like (6) separate in our data.

1.3. Prosodic features in the data

In our analysis, we will employ the prosodic description and transcription developed by Crystal (Crystal and Quirk (1964); Crystal (1969)); the reasons for this are partly of a practical nature, as our data have been transcribed using Crystal's notation system (see below). The basic unit for prosodic analysis is the Tone Unit (TU for short). TU's consist of a Nucleus (the most prominent syllable), which may be preceded by a Head (running from the first pitch-prominent syllable – i.e. the Onset – up to, but not including the Nucleus) and a Pre-head (everything coming before the Onset), and which may be followed by a Tail (carrying on the pitch movement of the Nucleus, in most cases). The Nucleus is the only obligatory element in the TU; the other elements are optional.

There are three types of rising Nuclei in Crystal's classification. The first one is the simple *Rise*, with a rising pitch from the Nucleus onwards

(transcribed as 'x̌'). The second tone is the complex *Fall–Rise*, with a changing pitch movement. Finally, there is a compound tone, the *Fall+Rise*, which has two separate Nuclei, one with a falling, one with a rising pitch (the latter two tones are transcribed as 'x̌' and 'x̀x̌', respectively). This study investigates all three types of Rising tone; the term *Rising Tone* is employed as a cover-term.

Pitch range is marked in two different ways. First of all, the pitch range of the Nucleus (and the Onset) is marked as normal (not marked in the transcription), Narrow ('*N*'), or Wide ('*W*'); in addition, the starting-point of the pitch movement can be normal (not marked), High ('*H*'), Extra-high ('*E*'), or Low ('*L*'). A Nucleus marked as '*HW*', for instance, starts high in the pitch range and has an end-point which is relatively far removed from its starting-point.

Secondly, the pitch range of the remainder of the TU (especially the Head) is captured by the so-called Booster system (Crystal (1969)). Normally, each stressed syllable has a slight drop in pitch compared to the preceding stressed syllable; this is the unmarked case. However, other possibilities exist. A syllable can be very high ('!!'), higher than the preceding pitch-prominent syllable ('!'), higher than the preceding syllable (':'), at the same level as the preceding syllable (this is labelled 'continuance') ('·'), low in pitch (','), or very low in pitch (';'). The latter two possibilities are lower than the unmarked case (the slight drop), the other possibilities are higher. This system makes it possible to follow the pitch movement throughout the Head.

Apart from pitch movement and pitch range, we are also interested in occurrence of Pauses in the data. These are transcribed as '.' (very short pause), '-' (short), '–' (longer), or '—' (very long), respectively. Unfortunately, no precise durations in milliseconds are available for our data.

Other transcription conventions used in the data are the following: simultaneous speech (*...* or **...**); Onset (/); TU boundary (#); non-nuclear stresses (' or ''); subordinate TU's (square brackets); speaker identity (capitals); untranscribable syllables ('((syll))'). Loudness, tempo and paralinguistic features such as creaky voice have been left out of the transcription, since we are only concerned with Intonation, Pitch Range, and Pausal features.

The database consists of 9 files from the Survey of English Usage, based at University College, London.[2] All files represent surreptitiously recorded, spontaneous conversation (reference numbers accompanying the examples refer to the entries in Svartvik and Quirk (1980), where slightly simplified versions of the examples can be found). The corpus consists of a 45,000 word sample, or about 11,000 TU's; this yielded 156 inversion-interrogatives, and 60 queclaratives.

[2] We are indebted to Sidney Greenbaum, Director of the Survey of English Usage, for giving us access to the material used in this study. Survey files employed are the following: S.1.1, 1.2, 1.4, 1.5, 1.6, 1.8, 3.2, 3.3, 3.4.

2. Prosodic features of inversion-interrogatives

2.1. Intonation in inversion-interrogatives

In our investigation of Nuclear tones, we will start out by working on a formal basis, and thus including interrogatives without question status in our discussion. This allows us to check whether their intonation differs significantly from 'real' inversion-questions.

We are interested, first and foremost, in the final pitch movement of the interrogatives. This, however, poses a methodological problem; consider (7):

(7) A: /does he 'dò it all # be/fore the 'term *stárts # ((for the /whole))*
B: *I expect it's the .* kind of material he gets (S.1.4.699)

The Rise on *starts* is the final Nucleus of the interrogative; however, the hearer has apparently decided on the Question status of A's utterance before hearing this Rise, as he begins to formulate an Answer while A is still talking. The relevant tone, therefore, from our point of view, is the Fall on *do*. Every time such overlapping of Question and Answer occurs in the data, we have taken the final tone to be the Nucleus occurring before the overlapping sets in.

Table 1a shows the relative frequency of Rising versus Non-Rising tones on interrogatives; the latter include Falls (transcribed '$\grave{x}$'), Rise-Falls ('$\hat{x}$'), and

Table 1a
Nuclear tones on inversion-interrogatives.

	Non-Rising	Rising	Total
Genuine questions	51	68	119
Non-questions	23	14	37
(-rhetorical)	(7)	(4)	(11)
(-indirect SA)	(16)	(10)	(26)
Total no. of interr.	74	82	156
Total no. in corpus	7460	3103	10,563

Table 1b
Rising tones on inversion-interrogatives.

	Rise	Fall–Rise	Fall + Rise
Genuine questions	44	13	11
Non-questions	9	2	3
(-rhetorical)	(3)	(1)	(0)
(-indirect SA)	(6)	(1)	(3)
Total no. of interr.	53	15	14
Total no. in corpus	1242	1094	767

Level tones ('x̄'). Rising tones occur on 52.5% of the data; our first observation, then, must be that rising intonation is not really more frequent on interrogatives than non-rising intonation. However, Rising tones are significantly more frequent on inversion-interrogatives than they are in the corpus as a whole (only 29.4% of the Nuclei in the data are Rising).

Table 1b shows the results for Rising tones, split up into three different types of Nuclei. Note that Rises (i.e. excluding Fall–Rises and Fall + Rises) occur in a mere 34.0% of our interrogative data (53 instances). Although, once again, this frequency is significantly higher than the frequency in the total corpus (which is 11.8%), it does mean that, in absolute terms, Rises are less frequent in the data than are Falls (Level tones and Rise–Falls being rare).

The obvious next step is to see whether Rising tones are used to distinguish genuine inversion-questions from inversion-interrogatives without question-status (or 'non-questions'). In table 1a, we can see that this is not the case: Although the frequency of Rising tones is slightly higher in questions than it is in non-questions (57% v. 38%), this difference turns out to be insignificant.[3] The majority of interrogatives in the data do indeed have question status.

Non-questions can be split into two categories: 'Rhetorical questions' and 'Indirect speech acts'. The former (11 instances) are not followed by an Answer, be it a Marked or an Unmarked one; some of them, such as (8), are not even turn-final:

(8) A: /and he stayed ((did he)) . /*H*stay ((at)) Òxford to do a 'post 'grad yéar # or /*H*did he 'come immèdiately 'here # I can't remember he's working for a PhD here I think (S.1.6.63)

Others are turn-final, but simply express 'surprise':

(9) B: Mace has a link with Imola
A: /hás he # . /mm̀h # (S.1.2.602)

The remaining 26 non-questions are Indirect speech acts, mostly Requests, as in (10):

(10) B: *oh yes ((have you /got some mìlk for mé #))*
A: *-and* then he sent a message (S.1.4.35)

[3] χ^2-tests were used to calculate significance levels for all the data; the significance level is set at 0.01 throughout.

The non-question status of (10) is confirmed by the fact that there is no Answer. However, as we have pointed out in section 1.2, there may be a verbal response:

(11) A: /would you !stick that in his !pìgeonhóle #.
B: yès # (S.3.2.1088)

We have argued that instances such as (11) are Requests with a Question element in them. The intonation of these 'Requests+Questions', however, does not differ significantly from the one used on 'normal' Requests such as (10); we have therefore simply grouped them together in table 1a. Other speech acts, such as Invitations, also occur, albeit less frequently:

(12) A: /aren't you 'going to sit dówn # -
B: /*H*mhm̂ # - (S.1.8.150)

Note, once again, the presence of a verbal response.

Turning now to the genuine inversion-questions in our corpus, we see that non-rising tones are nearly as frequent as rising tones (51 v. 68 instances; table 1a). As far as the Rising tones are concerned (table 1b), Rises are most common (36.9%), more so than Fall–Rises (10.9%) and Fall+Rises (9.3%). On the whole, Falls (e.g. (13)) are more common than Rises (e.g. (14)):

(13) A: well # does "/that mean that they insisted on :dealing with :yòu #
B: yeàh . mhm̀ # (S.1.2.564)
(14) A: is /this a spare páper #
B: yeah I actually got it for you (S.1.1.32)

(13) and (14) are good examples of Unmarked Answers; an instance of a more Marked Answer is (15):

(15) A: - /do you !like 'this 'work hére . # in /this depártment # -
B: you were here were you once - (S.1.6.132),

where B, in a very roundabout way, confirms A's question.

We have argued that strict adjacency is not always necessary, and this is borne out by questions such as (16) and (17), which resemble (4) and (5), respectively:

(16) B: /is there !anything that'd be - worth his ,*W*whíle 'reading # -
A: /is it . for this sŭmmer # . *he*
B: *yeah* he's taking finals for the third time this summer -
A: well there are a number of things (S.1.4.899)

(17) C: ((/wéll # . have we de/cìded thén #)) . the /grand !tòur #
B: yès (S.1.5.691)

Instances such as these are relatively rare, though.

One special type of inversion-question needs mentioning here, viz. 'Alternative Questions'. In this type of polar question, the speaker in a way combines two polar questions, by offering two alternatives; rejection of one alternative implies acceptance of the other one. The typical Intonation pattern for Alternative Questions consists of a Rise (or Fall–Rise) on the first part, followed by a Fall on the second part. However, a combination of two Falls is also possible, as in (18):

(18) /are you . [àl!] 'members of a re'search . :pròject # - or just a gròup # (S.1.5.418)

Summing up this section, we can conclude that Rising intonation is certainly not a necessary feature of inversion-questions. More importantly, on a quantitative basis, it does not make sense to regard Rises, or even Rising intonation in general, as being the 'normal' tone on inversion-questions: Rises, though relatively frequent, occur in less than 37% of the data. Unfortunately, the nature of our data does not allow us to check the attitudinal implications of the various tones; Rising nuclei might be judged as being more 'neutral' than non-rising ones. The only way to verify this is through experimental research. However, a word of caution is in order here, as in some cases the attitudinal overtone of an utterance depends more on the choice of lexical material than on the Intonation pattern. For instance, utterances such as (19) are likely to be interpreted as 'emotive' rather than 'neutral', irrespective of the type of Nucleus:

(19) Are you out of your bleedin' mind?

2.2. *Pitch range in inversion-interrogatives*

Three different aspects of pitch range will be discussed in this section. First of all, we will look at the pitch range of the Nucleus. The Nucleus can be Wide or Narrow in range, and can have a High or Low starting point. On the whole, however, special pitch range features are very rare in our data, apart from a slight tendency for Rises to be Wide and/or High. There are no significant differences as regards the nuclear range of Questions as opposed to Non-questions.

Much more interesting is the pitch range of the Head, represented by the 'booster' system (table 2). There is a marked tendency for questions ending in a Falling tone to have one or more high boosters within the Head, i.e. to have

a Head which steps up in pitch; this is not true to the same extent for non-questions ending in a Falling tone. Rising questions tend to have a low booster (i.e. a down-stepping Head) rather than a high one, but both are relatively rare.

Table 2
Pitch range of Head in inversion-interrogatives.

	high booster (!!/!/:)		low booster (,/;)	
	[+Q]	[−Q]	[+Q]	[−Q]
Falling	25	9	1	–
Rise	5	–	9	3
Fall–Rise	2	–	1	–
Fall + Rise	–	–	–	–

Thirdly, and finally, one can look at the pitch range of the interrogative 'as a whole', i.e. in relation to surrounding utterances. There seems to be a tendency for question-TU's to be situated High in the speaker's pitch range, irrespective of whether a Falling or a Rising tone is being used.

Regrettably, it is impossible to evaluate the precise impact of pitch range in our data. The only thing worth remembering at this stage appears to be the tendency for 'Falling' questions to have a rising Head.

2.3. *Pausal aspects of inversion-interrogatives*

Although a lot of questions end on a non-rising Nucleus, one might argue that Rising Nuclei make it easier for the hearer to recognize an utterance as being a question. If this is indeed the case, this should be reflected in the time it takes him/her to react with a response; in other words, it is worth checking the occurrence of pauses between Question and Answer.

Table 3
Pausal features in inversion-questions.

	Falling	Rising	(Rise)
No pause between Q and A	25	40	(23)
Pause between Q and A	23	28	(21)

Table 3 correlates nuclear tone types with the presence/absence of pauses. The figures speak for themselves: There are no significant differences between Falling and Rising Nuclei, as far as presence or absence of Pauses is concerned; the same is true if we only take simple Rises into account. More than half of the Falling questions have no pause at all; in fact, Question and

Answer sometimes partially overlap. When a pause does occur, it is mostly either a very short ('.') or a short one ('-'). The same holds for Rising questions: More than 50% are not followed by a pause.

The presence of a pause, and its length, is of course dependent on other factors as well, such as for instance the complexity of the Question. However, all other things being equal, we can conclude that the choice of nuclear tone does not appear to have any impact on the time it takes hearers to respond to an inversion-question. This suggests that the importance of Intonation for question recognition is overestimated.

3. Prosodic features of Queclaratives

3.1. Intonation in queclaratives

If Rising intonation is at all important in polar questions, this ought to be especially so in Queclaratives. Since there are no syntactic cues to suggest that a declarative utterance might have question status, and hence be a Queclarative, one might be tempted into the prediction that Rising intonation does the job of 'adding' question status to a declarative. However, previous experimental research (Geluykens (1986)) has shown that intonation by itself cannot turn a declarative into a Queclarative; pragmatic cues are much more important for determining the statement/question status of an utterance. Only when other cues are kept 'neutral' does intonation begin to have some influence. There is all the more reason, then, to assume that in conversation, where contextual–pragmatic features are far more elaborate, intonation plays only a minor role. More specifically, we predict that Rising intonation is irrelevant in Queclaratives. Previous corpus-based research (admittedly, on Edinburgh Scottish English) carried out by Brown et al. (1980) has already shown that Queclaratives often end in a 'fall-to-low'; we will investigate whether this is also true in 'standard' Southern English.

Our question recognition procedure, as represented in (3c), only accepts Unmarked responses to Queclaratives; by thus excluding all potential 'Marked' responses, we can be reasonably sure that the 60 remaining utterances, on which this section is based, are indeed Queclaratives. It is often stated that queclaratives expect a 'positive' response, i.e. the speaker expects the Answer to confirm the proposition expressed in his/her Question; this leads Stenström to labelling them 'Q-confirm' (Stenström (1984)). It must be pointed out, however, that, although most Queclaratives indeed have a confirmative Answer, this need not be the case; e.g.:

(20) B: and /you're on thàt #
A: nò no no . Dave is . (S.1.2.1386)

Table 4a
Nuclear tones on queclaratives.

	Non-Rising	Rising	Total
Queclaratives	41	19	60
Total corpus	7460	3103	10,563

Table 4b
Rising tones on queclaratives.

	Rise	Fall–Rise	Fall + Rise
Queclaratives	6	6	7
Total corpus	1242	1094	767

This type of response is somehow more Marked than a confirmative one (hence the Rise–Fall on *no* and the repetitions). Some responses which are negative in form are nevertheless clearly confirmations, as in:

(21) B: you /don't mean the :Hussey 'thingummy *and :whàtsit #*
A: nó (S.3.2.826)

Answering *yes* here would imply that A *does* indeed mean the proposition expressed by B; since the Queclarative is negative in form, the confirmative Answer must follow the polarity of the Question. In what follows, we will not be concerned with the precise nature of the Queclarative responses, but concentrate on the Queclaratives themselves.

Table 4 shows the relative frequency of Rising and Non-Rising tones, compared with their frequency in the database as a whole (taking into account overlapping of Question and Answer). It will be observed that the majority of Queclaratives have a non-rising Nucleus, mostly a Fall (57% of the data); this is comparable to the overall frequency of Falls in the corpus (64%). In fact, there are no significant differences between the relative tone frequency in Queclaratives, and the frequency found in the corpus a a whole. This not only confirms our initial claim that Rising tones are not the 'normal' tones used on Queclaratives; it also suggests that intonation is virtually irrelevant as a question-cue. Examples of 'Falling' Queclaratives are (20) and (21) above, and (22) to (25) below:

(22) A: /Steven Peel sup!pòrted you # .
B: yès . most curious (S.1.2.85)
(23) B: ((then)) this /isn't your – ((:chòre)) - #
A: nô . (S.1.2.1333)
(24) C: I don't think you've met :Nelly 'Cartwright up:stàirs #
A: nò (S.1.5.792)

(25) D: but . /you en!jòyed it # *first of àll* #[4]
A: *oh yês* (S.1.5.1283)

Note that all responses are confirmative.

As for Rising tones, all three types are about equally frequent (table 4b): 10% are simple Rises (e.g. (26)), 10% are Fall–Rises (e.g. (27)), 12% are Fall+Rises (e.g. (28)):

(26) A: /you are now !formulating a plán # *- to*
B: *I àm . yès* (S.1.2.574)
(27) B: /they're sort of !piěces #
A: yès (S.1.8.183)
(28) C: I /don't think you've bèen up'stairs yét #
A: m no only just to the loo (S.1.5.730)

The figures for Rising tones differ a lot from those found for inversion questions, where Rises are far more frequent than the other two types of Rising tone.

3.2. *Pitch range in queclaratives*

Of the three pitch range features marked in the data (pitch range of the Nucleus, of the Head, and of the TU as a whole), only the pitch range of the Head is worth mentioning; figures can be found in table 5.

There is a very high frequency of 'high boosters' – i.e. Heads stepping up in pitch – accompanying Falls in Queclaratives: 82.3% of them (28 instances) have one or more up-stepping syllables in the Head (mostly ':' (24) or '!' (22), but '!!' also occurs). In quite a number of instances, this step-up in pitch occurs immediately before the Nucleus (this is the case in instances (22) to (24)); this makes the Nucleus more or less equivalent to a High Nucleus, since the starting-point of the Fall will obviously be rather High in the pitch range.

Table 5
Pitch range of Head in queclaratives.

	High booster	Low booster
Falling tone	28	2
Rise	–	–
Fall–Rise	4	–
Fall+Rise	–	–

[4] Note that such utterances are already, by the nature of their lexical material, 'question-prone' (Geluykens (1987)).

If there is such a thing as a 'normal' intonation pattern for Queclaratives, this combination of a nuclear Fall preceded by a rising Head seems the most likely candidate. It will be recalled that this pattern is also relatively frequent on Falling inversion questions, though less outspokenly so.

High boosters also occur in four out of six Fall–Rises in the data (table 5). The implications are similar as for Falls: As a result of the rising Head, the falling part of the Fall–Rise will start rather High in the range (e.g. (27)).

Two of the Rises in the data have a High Nucleus, and in two other instances the whole TU is High in the pitch range. This means that, in two thirds of the data, the end-point of the Rise is relatively High.

3.3. *Pausal features in queclaratives*

Table 6 clearly shows that the choice of a Rising or Falling tone does not have any influence on the occurrence of pauses between Question and Answer: There are no significant differences between presence/absence of pauses in Falling versus Rising tones. In fact, Rising tones are more likely to be followed by a pause than are Falling tones (or Level tones, for that matter). The majority of pauses in the data, both for Falling and for Rising tones, are either short ('-' or very short ('.').

Table 6
Pausal features in queclaratives.

	Falling	Rising	(Rise)
Pause absent	24	9	(2)
Pause present	10	10	(4)

Put differently: The choice of nuclear tone does not appear to have any influence on the time it takes hearers to recognize an utterance as being a Queclarative.

4. Conclusions and further perspectives

4.1. *Prosody and polar questions*

Our findings in sections 2 and 3, tentative as they may be in some respects, justify one major conclusion, viz. that the role of Rising intonation in polar questions is overrated. Rising tones – and, more specifically, Rises – are by no means as frequent as one might expect.

In Inversion questions, Rising tones are more frequent than they are in the data as a whole; this is especially true for simple Rises, more so than for Fall–Rises and Fall+Rises. However, in absolute terms, the most frequent

type of Nucleus is a Falling one (mostly a simple Fall, rarely a Rise–Fall). Moreover, Rising intonation is *not* used for distinguishing genuine Inversion questions from Inversion interrogatives without question status (tables 1a and 1b). Neither is it the case that questions ending with a rising pitch are easier to decode by the hearer, judging by the occurrence of Pauses in the data (table 3). This suggests that, even if Rising tones do have a function in questions, this function definitely cannot be the marking of their question status. Falling tones on Inversion questions are often preceded by a rising Head, a feature which certainly deserves more attention than we can give it here.

The most striking outcome of our analysis is the very low frequency of Rising tones in Queclaratives (table 4a–b). Falling tones are far more frequent; in the majority of cases, these are preceded by a 'high booster' (table 5), i.e. a step-up in pitch in the Head of the TU. Once again, judging by the presence/absence of Pauses in the data (table 6), intonation does not seem to have an effect on the recognition of questions. Since prosody does not appear to be relevant, the most likely alternative must be that Queclaratives acquire question status by virtue of the context in which they appear. As we have already pointed out elsewhere (Geluykens (1986)), the lexical material might give the hearer important clues as to whether a declarative utterance is a statement or a Queclarative; for instance, the use of *you don't mean* by the speaker in (21) makes the utterance decidedly more 'question-prone', irrespective of the intonation contour.

If Rising intonation is not used for signalling questions, what *is* it used for in conversation? We will try to shed more light on this problem in the following section.

4.2. Uses of Rising intonation in conversation

One way of finding out why speakers use Rising tones, is to check the discourse locations in which Rising intonation occurs. Whereas in sections 2 and 3 we have taken 'syntax' as a starting-point, and tried to relate it to Intonation, we will employ the opposite strategy here: we will start with Rising intonation, and relate it to a non-prosodic aspect, viz. place of occurrence in conversational discourse.

Results can be found in table 7; we will discuss them in some detail here. First of all, a threefold distinction is made into Fillers (items such as *you know, I mean*, and the like), Non-clause-final Nuclei, and Clause-final Nuclei. The latter are split up into Turn-final and Non-turn-final ones. Turn-final rising tones can be either on Questions – we have included questions which are, strictly speaking, not turn-final in this category – or on 'other' utterances. This should give us some idea as to the respective frequencies of Rising tones in these locations.

Table 7
Discourse locations of Rising tones.

	Rise	Fall–Rise	Fall + Rise
Fillers	246 (20.2%)	51 (4.7%)	28 (3.7%)
Non-clause-final	246 (20.2%)	302 (27.9%)	153 (20.4%)
Clause-final	728 (59.6%)	730 (67.4%)	571 (75.9%)
Non-turn-final	538 (44.1%)	620 (57.3%)	472 (62.7%)
Turn-final	190 (15.5%)	110 (10.1%)	99 (13.2%)
Polar Q.	50 (4.1%)	19 (1.7%)	18 (2.4%)
Other	140 (11.4%)	91 (8.4%)	81 (10.8%)
Total no. of tones	1220 (100%)	1083 (100%)	752 (100%)

It is clear that rising tones on polar questions are marginal: Grouping together Rises, Fall–Rises and Fall + Rises, Rising Nuclei very rarely occur on polar questions (2.85% of the rising tones). The most common discourse locations for rising tones are either non-finally in a clause (19.7% of the data for the three rising tones grouped together), or non-finally in a turn (53.4%). In other words, Rising Intonation serves as a 'non-finality marker', a function which is also mentioned by Cruttenden (1981) and Bolinger (1982, 1985). In conversation, marking the incompleteness of a turn is clearly important for the speaker. Every time the end of a clause is reached, there is a potential 'transition relevance place' (Sacks et al. (1974)) within the turn, where someone else may take the floor and begin a new turn. Rising intonation probably serves as a cue to signal to the hearer that the turn is not complete, and that the speaker does not wish to be interrupted.

Another frequent location of rising tones, more specifically Rises, is on Fillers (20.2%).

As for turn-final Rising tones (about 13% of the data), less than a quarter of them (i.e. 21.8% of the turn-final rising tones) are situated on polar questions. Put differently, whichever way one looks at the data, it is clear that Rising tones in general, and simple Rises in particular, are far from being the 'normal' tones used for asking polar questions. It is also striking that there is hardly any difference between Rises, Fall–Rises and Fall + Rises, as far as their discourse locations are concerned (table 7).

4.3. *Rising intonation on questions as a universal*

It is sometimes claimed in the literature that the frequent use of a Rising contour to mark polar questions is an intonational universal, especially in languages which have no 'syntactic' marking of questioning corresponding to Inversion, such as Portuguese. So Cruttenden states:

"The intonation of yes/no questions, whether co-occurring with morphological/syntactic marking or not, is almost invariably reported as having either a 'terminal rise' or in some way a higher pitch than the corresponding statement pattern" (Cruttenden (1986: 162)).

Indeed, both in Bolinger (1978) and Ultan (1978), it is claimed that the majority of languages employ these features for Questioning.

However, considering what is stated in the literature about English, and the way it contrasts with our findings here, it remains to be seen whether the same claims are still valid if one investigates the use of intonation in *actual conversational discourse* in a large sample of languages. Even in languages that lack morpho-syntactic marking of polar questions, it does not necessarily follow that the only way to mark an utterance as being a question is through intonation; if this were true, the same tendency would probably show up in English Queclaratives. Section 3 has shown that this is not the case. Perhaps a much more likely universal feature of Rising intonation is its signalling of 'non-finality' in the turn-taking system in conversation; the only way to find out is through analyzing large samples of conversation, as we have done here for English.

On the whole, it would be sensible to avoid over-simplification and over-generalization in describing the functions of intonation. It is to be hoped that 'wishful thinking' will be replaced by rigorous quantitative and experimental research as a method for discovering the uses of intonation in discourse.

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